

Qualification Pack



Junior Technician- Solar Cold Storage

QP Code: SGJ/Q4002

Version: 2.0

NSQF Level: 3

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SGJ/Q4002: Junior Technician- Solar Cold Storage

Brief Job Description

Junior Technician-Solar Cold Storage performs installation along with operations and maintenance and of a solar cold storage/cold room solution while meeting the performance and reliability needs of customers by incorporating quality craftsmanship and complying with all applicable industry standards, quality and safety requirements.

Personal Attributes

The individual should be punctual, attentive, sincere and should have an ability to bring a positive and helpful attitude to work. The individual shall also be highly motivated towards achieving an organizational goal so as to inspire others at work as well. The individual should also be knowledgeable enough to diagnose the cause of an issue with regards to system installation and fully skilled to figure out how to fix it.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4006: Basics of Solar cold storage system installation and its operation](#)
2. [SGJ/N0702: Site Survey and pre-requisites for installation of solar cold storage system](#)
3. [SGJ/N3901: Elements of solar cold storage system and their functions](#)
4. [SGJ/N0623: Technical aspects for installation, usage, performing O&M and cost of solar cold storage system](#)
5. [SGJ/N1301: Availing low- cost financing and Case studies showcasing best practices in system installation and operation of solar cold storage system](#)
6. [SGJ/N0801: Maintain Health and Safety at Solar cold storage project site](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician

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Country	India
NSQF Level	3
Credits	9
Aligned to NCO/ISCO/ISIC Code	NCO2015/7127.0100
Minimum Educational Qualification & Experience	10th grade pass with NA of experience OR 8th grade pass (or Equivalent) with 3 Years of experience relevant experience in Solar PV Project Helper OR 5th grade pass (or equivalent) with 6 Years of experience relevant experience in Solar PV Project Helper OR Previous relevant Qualification of NSQF Level (2) with 3 Years of experience relevant experience in Solar PV Project Helper
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	11/03/2029
NSQC Approval Date	12/03/2026
Version	2.0
Reference code on NQR	QG-03-GJ-05083-2025-V1-SCGJ
NQR Version	2

Remarks:

Total 330 Hours: (Theory: 90 hours+Practical:150 hours+ 30 hours of employability skills + 60 hours of OJT)

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SGJ/N4006: Basics of Solar cold storage system installation and its operation

Description

This unit provides overall concept of solar cold storage system with its functions and key components. This unit also provide basic skills and knowledge required to manage and operate cold storage system installation work.

Scope

The scope covers the following :

- key importance of cold storage for storing various agricultural/ horticulture produce
- key insights of solar cold storage system installation and operation

Elements and Performance Criteria

Solar cold storage system installation and its operation

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss solar energy and its various applications, including in solar cold storage.
- PC2.** Discuss the basic functions of a solar cold storage system and how that meet the requirements for storing agricultural/horticultural produce by farmers/Farmers producers' organisations (FPOs), etc at farm/Mandi level.
- PC3.** Discuss in brief the key components of the solar cold storage system and their functions
- PC4.** Discuss the overall concept of solar cold storage system, illustrate its schematic and explain the main functions of main components
- PC5.** Briefly explain the prospects, process, and practice of solar/ cold storage system in India
- PC6.** Discuss key drivers, trends, and overall opportunities offered by solar cold storage system to various stakeholders.
- PC7.** Discuss the importance of this job role and the basic skills, knowledge and responsibility of a technician in cold storage market space
- PC8.** Discuss the evolving landscape of Original Equipment Manufacturers (OEMs), system assemblers and developers in solar cold storage business space
- PC9.** Provide an overview of key stakeholders like Farmer Producer organisations (FPOs)/ Self Help Groups (SHGs) and operators in mandi etc
- PC10.** Explain concerned policies, schemes and regulations issued by the Central/State Government along with Standards/benchmarks associated with solar cold room (including cost), if any
- PC11.** Illustrate the significance of solar energy and its applications across various technologies including solar cold storage solution.
- PC12.** Show how solar cold storage solutions are useful for storing and extending the shelf life of agricultural and horticultural produce
- PC13.** Illustrate schematic of solar cold storage system and explain the functions of all key components

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- PC14.** Show how to analyse key drivers, trends, and overall opportunities solar cold storage business offers
- PC15.** Outline the advantages of solar cold storage system and show how it can be useful for various stakeholders
- PC16.** Show how basic skills and required knowledge can be inculcated to meet the overall responsibility of a Junior technician in solar cold storage market space.
- PC17.** Analyse the landscape of various OEMs, system assemblers and developers
- PC18.** Analyse the role of key stakeholders like FPOs/ other operators in Mandi etc
- PC19.** Outline concerned policies, regulations, guidelines etc issued by State/Central Governments along with relevant Standards or benchmarks associated with solar cold storage system

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Brief concept of solar cold storage system
- KU2.** Various application of Solar Cold Storage
- KU3.** Basic functions of solar cold storage system components
- KU4.** key trends and overall opportunities offered by solar cold storage system to various stakeholders
- KU5.** policies, schemes and regulations issued by the Central /State Government
- KU6.** role of key stakeholders like FPOs/ Self Help Groups (SHGs) other operators in Mandi
- KU7.** basic skills, knowledge and responsibility of a technician in cold storage market space

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and understand manuals, health and safety instructions, guidelines memos, other company documents
- GS3.** read vernacular/English language
- GS4.** follow organizational rule-based decision making process
- GS5.** respond appropriately to any queries by customers and other stakeholders
- GS6.** read and understand manuals, health and safety instructions, guidelines memos, other company documents
- GS7.** behave politely and appropriately with all
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization
- GS10.** take decision with systematic course of actions and/or response

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Solar cold storage system installation and its operation</i>	24	26	-	-
PC1. Discuss solar energy and its various applications, including in solar cold storage.	3	-	-	-
PC2. Discuss the basic functions of a solar cold storage system and how that meet the requirements for storing agricultural/horticultural produce by farmers/Farmers producers' organisations (FPOs), etc at farm/Mandi level.	3	-	-	-
PC3. Discuss in brief the key components of the solar cold storage system and their functions	3	-	-	-
PC4. Discuss the overall concept of solar cold storage system, illustrate its schematic and explain the main functions of main components	2	-	-	-
PC5. Briefly explain the prospects, process, and practice of solar/ cold storage system in India	2	-	-	-
PC6. Discuss key drivers, trends, and overall opportunities offered by solar cold storage system to various stakeholders.	2	-	-	-
PC7. Discuss the importance of this job role and the basic skills, knowledge and responsibility of a technician in cold storage market space	3	-	-	-
PC8. Discuss the evolving landscape of Original Equipment Manufacturers (OEMs), system assemblers and developers in solar cold storage business space	2	-	-	-
PC9. Provide an overview of key stakeholders like Farmer Producer organisations (FPOs)/ Self Help Groups (SHGs) and operators in mandi etc	2	-	-	-
PC10. Explain concerned policies, schemes and regulations issued by the Central/State Government along with Standards/benchmarks associated with solar cold room (including cost), if any	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Illustrate the significance of solar energy and its applications across various technologies including solar cold storage solution.	-	3	-	-
PC12. Show how solar cold storage solutions are useful for storing and extending the shelf life of agricultural and horticultural produce	-	2	-	-
PC13. Illustrate schematic of solar cold storage system and explain the functions of all key components	-	3	-	-
PC14. Show how to analyse key drivers, trends, and overall opportunities solar cold storage business offers	-	3	-	-
PC15. Outline the advantages of solar cold storage system and show how it can be useful for various stakeholders	-	2	-	-
PC16. Show how basic skills and required knowledge can be inculcated to meet the overall responsibility of a Junior technician in solar cold storage market space.	-	2	-	-
PC17. Analyse the landscape of various OEMs, system assemblers and developers	-	3	-	-
PC18. Analyse the role of key stakeholders like FPOs/ other operators in Mandi etc	-	4	-	-
PC19. Outline concerned policies, regulations, guidelines etc issued by State/Central Governments along with relevant Standards or benchmarks associated with solar cold storage system	-	4	-	-
NOS Total	24	26	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4006
NOS Name	Basics of Solar cold storage system installation and its operation
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	3
Credits	1
Version	2.0
Last Reviewed Date	12/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	12/03/2026

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SGJ/N0702: Site Survey and pre-requisites for installation of solar cold storage system

Description

This unit is about analyzing parameters and other pre-requisites to be used in site survey for installation of solar cold storage system. This unit also provides an overview of design considerations and handling of tools/ instruments used in site survey and system installation.

Scope

The scope covers the following :

- analysis of various parameters for installation of solar cold storage system
- survey site for installation of solar cold storage system
- Discuss basics of design considerations for solar cold storage system
- Discuss pre-requisites for installing solar cold storage system
- Describe various tools used in site survey and system installation

Elements and Performance Criteria

Analysis of parameters used in installation of solar cold storage system

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss various parameters for system installation, any design considerations, create suitable checklist and explain how to identify the location where solar cold storage system shall be installed
- PC2.** Discuss how to analyse key information (e.g. quantity/type of produce, distance from mandi etc) for identifying the sites for installation of solar cold storage
- PC3.** Discuss how to assess the potential and demand for solar cold storage solution
- PC4.** Discuss how to map the power requirement, assess system demand/ sizing/ usage and meet supply provisions (from solar/ Grid/ DG etc) for cold storage system installation and its successful operation
- PC5.** Discuss how to use various tools/instruments for site survey.
- PC6.** Discuss the requirement for any civil work required for solar cold storage system installation
- PC7.** Illustrate and Analyse various parameters including any design considerations for siting the location for installing solar cold storage system.
- PC8.** Show how to access required data and analyse key info for performing site survey and create checklist for installing solar cold storage solution.
- PC9.** Demonstrate how to perform calculations for estimating demand/potential for solar cold storage in a given area
- PC10.** Illustrate how to map the power requirement, assess system demand/ sizing and usage and meet supply provisions (from solar/ Grid/ DG etc) for cold storage system installation and its successful operation
- PC11.** Show how to perform site survey by using various tools/instruments.
- PC12.** Show how to perform the required civil work for system installation, if any

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** various parameters for solar cold storage system installation
- KU2.** government/corporate policies and guidelines on solar cold storage installation
- KU3.** concerned legislative, organizational, site requirements and related procedures
- KU4.** different types of tools used for system installation
- KU5.** assess the potential and demand for solar cold storage solution

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** effective communication with regards to installation and related instructions to team
- GS2.** planning and organization of work to meet deadlines
- GS3.** apply domain knowledge, observations and data to select course of action to perform tasks related to solar cold storage system
- GS4.** choose best methods to complete assigned tasks
- GS5.** solve problems effectively
- GS6.** manage time efficiently

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analysis of parameters used in installation of solar cold storage system</i>	25	25	-	-
PC1. Discuss various parameters for system installation, any design considerations, create suitable checklist and explain how to identify the location where solar cold storage system shall be installed	4	-	-	-
PC2. Discuss how to analyse key information (e.g. quantity/type of produce, distance from mandi etc) for identifying the sites for installation of solar cold storage	4	-	-	-
PC3. Discuss how to assess the potential and demand for solar cold storage solution	4	-	-	-
PC4. Discuss how to map the power requirement, assess system demand/ sizing/ usage and meet supply provisions (from solar/ Grid/ DG etc) for cold storage system installation and its successful operation	4	-	-	-
PC5. Discuss how to use various tools/instruments for site survey.	5	-	-	-
PC6. Discuss the requirement for any civil work required for solar cold storage system installation	4	-	-	-
PC7. Illustrate and Analyse various parameters including any design considerations for siting the location for installing solar cold storage system.	-	5	-	-
PC8. Show how to access required data and analyse key info for performing site survey and create checklist for installing solar cold storage solution.	-	4	-	-
PC9. Demonstrate how to perform calculations for estimating demand/potential for solar cold storage in a given area	-	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Illustrate how to map the power requirement, assess system demand/ sizing and usage and meet supply provisions (from solar/ Grid/ DG etc) for cold storage system installation and its successful operation	-	4	-	-
PC11. Show how to perform site survey by using various tools/instruments.	-	4	-	-
PC12. Show how to perform the required civil work for system installation, if any	-	4	-	-
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0702
NOS Name	Site Survey and pre-requisites for installation of solar cold storage system
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Technician
NSQF Level	3
Credits	1
Version	2.0
Last Reviewed Date	12/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	12/03/2026

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SGJ/N3901: Elements of solar cold storage system and their functions

Description

This unit outlines the key elements of solar cold storage system, their functions as well as functions of all key components.

Scope

The scope covers the following :

- Describe the key elements of solar cold storage system and their functions
- Discuss functions of all key components of the system

Elements and Performance Criteria

key elements and functions of solar cold storage system

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the detailed prospects, concept and practice of utilisation of solar cold storage system in India
- PC2.** Discuss how to perform planning for installation of solar cold storage solution
- PC3.** Discuss how “plug and play” modular, solar-powered cold room is designed for 24/7 storage, thereby addressing the problem of post- harvest losses in fruits, vegetables and other perishable food.
- PC4.** Discuss how to identify various tools and tackles and utilise them for various assembly and installation functions
- PC5.** Discuss key elements of a solar cold storage unit, their specifications and overall installation of the system
- PC6.** Explain basic design and fabrication of solar cold storage system along with its key components (e.g. refrigeration system, etc)
- PC7.** Discuss any pre requisites for the overall system installation, as per the site requirements.
- PC8.** Discuss key Information on storage/ cooling requirements of different fruits, vegetables and other farm products [e.g. shelf life, temperature range, humidity etc.]
- PC9.** Discuss how cooling requirement during the evening hours can be met from the thermal backup system
- PC10.** Demonstrate the functions of all key components of the solar cold storage system and showcase its successful operation for storing a range of agricultural/horticultural produce
- PC11.** Show how to use various tools and tackles safely for installation the system and all its components
- PC12.** Perform basic design and fabrication of solar cold storage system including of key components like refrigeration, thermal energy system etc
- PC13.** Analyse various specifications of the key components of solar cold storage system and how to perform system assembly/ installation
- PC14.** Analyse any pre- requisites for the overall system installation, as per the site requirements.

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- PC15.** Analyse key information on storage/cooling requirements of various produce (e.g. temperature, humidity, shelf life etc)
- PC16.** Show how cooling requirement during the evening hours can be met from the thermal backup system.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the detailed prospects, concept and practice of utilisation of solar cold storage system in India
- KU2.** various tools and tackles safely for installation the system
- KU3.** efficiency, cost and typical specifications along with functioning and operating principle of different components of the system
- KU4.** organisational culture and customer support policy
- KU5.** general terminologies associated with solar cold storage solutions

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read English and/or vernacular language
- GS2.** respond appropriately to any queries by customers and other stakeholders
- GS3.** follow organisational code of conduct
- GS4.** read health and safety instructions and signage
- GS5.** work constructively and collaboratively with others
- GS6.** make timely decisions for efficient utilization of resources

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>key elements and functions of solar cold storage system</i>	25	25	-	-
PC1. Explain the detailed prospects, concept and practice of utilisation of solar cold storage system in India	3	-	-	-
PC2. Discuss how to perform planning for installation of solar cold storage solution	3	-	-	-
PC3. Discuss how “plug and play” modular, solar-powered cold room is designed for 24/7 storage, thereby addressing the problem of post-harvest losses in fruits, vegetables and other perishable food.	3	-	-	-
PC4. Discuss how to identify various tools and tackles and utilise them for various assembly and installation functions	3	-	-	-
PC5. Discuss key elements of a solar cold storage unit, their specifications and overall installation of the system	3	-	-	-
PC6. Explain basic design and fabrication of solar cold storage system along with its key components (e.g. refrigeration system, etc)	3	-	-	-
PC7. Discuss any pre requisites for the overall system installation, as per the site requirements.	3	-	-	-
PC8. Discuss key Information on storage/ cooling requirements of different fruits, vegetables and other farm products [e.g. shelf life, temperature range, humidity etc.]	2	-	-	-
PC9. Discuss how cooling requirement during the evening hours can be met from the thermal backup system	2	-	-	-
PC10. Demonstrate the functions of all key components of the solar cold storage system and showcase its successful operation for storing a range of agricultural/horticultural produce	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Show how to use various tools and tackles safely for installation the system and all its components	-	3	-	-
PC12. Perform basic design and fabrication of solar cold storage system including of key components like refrigeration, thermal energy system etc	-	3	-	-
PC13. Analyse various specifications of the key components of solar cold storage system and how to perform system assembly/ installation	-	4	-	-
PC14. Analyse any pre- requisites for the overall system installation, as per the site requirements.	-	4	-	-
PC15. Analyse key information on storage/cooling requirements of various produce (e.g. temperature, humidity, shelf life etc)	-	4	-	-
PC16. Show how cooling requirement during the evening hours can be met from the thermal backup system.	-	4	-	-
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N3901
NOS Name	Elements of solar cold storage system and their functions
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	3
Credits	2
Version	2.0
Last Reviewed Date	12/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	12/03/2026

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SGJ/N0623: Technical aspects for installation, usage, performing O&M and cost of solar cold storage system

Description

This unit describes the technical parameters of all key components of solar cold storage system, design, installation and operation & maintenance and how to analyse them.

Scope

The scope covers the following :

- Discuss the technical parameters of all key components of solar cold storage system
- Discuss key parameters for solar cold storage system design, installation and O&M

Elements and Performance Criteria

technical parameters of all key components of solar cold storage system

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss key technical aspects of system components, their specifications and step by step process for installation including of the following :Solar PV system, Refrigeration system, Insulation type/Thermal Energy Storage system, Digital configuration of Remote monitoring system
- PC2.** Discuss schematic and wiring diagrams along with Check list for O&M including preventive maintenance and routine check up and system monitoring, as provided in the product manual by the manufacturer/ developer/ System Assembler.
- PC3.** Describe the importance of conducting testing of all solar PV components and performing fault finding and analysis, continuity checks, polarity check etc.
- PC4.** Discuss all key aspects of system sizing, components assembly and overall installation and O&M
- PC5.** Describe the DO's and Don'ts of components/system installation, operation, material handling at site, system O&M, monitoring etc
- PC6.** Describe the importance of Earthing for the protection of solar PV system.
- PC7.** Explain how to perform routine maintenance activities required for each component of the solar cold storage system
- PC8.** Explain how to prepare and implement preventive maintenance schedule and reactive maintenance activities.
- PC9.** Explain the typical faults, their causes and resolution for all system components.
- PC10.** Explain the de-mounting of a solar PV power plant (after commissioning).
- PC11.** Discuss how to identify basic cost parameters for various solar cold storage system components, installation and O&M.
- PC12.** Discuss key aspects of system components warranty, Annual Maintenance Contract (AMC), service schedule, etc
- PC13.** Show how to analyse system components, their design attributes and specifications

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- PC14.** Demonstrate how to choose design criteria of solar cold storage as per assessment of site conditions
- PC15.** Demonstrate step by step process for installation of the system and its components, including for: Solar PV system, Refrigeration system, Insulation type/Thermal Energy Storage system, Digital configuration of Remote monitoring system
- PC16.** Show how to analyse schematic and wiring diagrams and ensure checklist for system usage, O&M and monitoring as per the technical manual provided by the manufacturer/ developer/ System Assembler.
- PC17.** Analyse all key aspects of system sizing, components assembly and overall installation and O&M
- PC18.** Describe the DO's and Don'ts of components/system installation, operation, material handling at site, system O&M, monitoring etc.
- PC19.** Demonstrate how to perform earthing for the protection of solar PV system
- PC20.** Demonstrate how to conduct testing of all solar PV components and performing fault finding and analysis, continuity checks, polarity check etc.
- PC21.** Demonstrate how to carry out service and routine maintenance work for each component.
- PC22.** Demonstrate how to prepare and execute preventive maintenance schedule and reactive maintenance activities
- PC23.** Demonstrate how to identify typical faults, their causes and resolution for all components
- PC24.** Demonstrate demounting of solar PV power plant
- PC25.** Show how to analyse basic cost parameters for solar cold storage system components, system installation and O&M.
- PC26.** Show how to analyse all key aspects of availing system components warranty, performing Annual Maintenance Contract (AMC), service schedule etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** technical parameters of all key components of solar cold storage system
- KU2.** key parameters for solar cold storage system design
- KU3.** key parameters for installation of solar cold storage system
- KU4.** key parameters for operation & maintenance of solar cold storage system
- KU5.** Characteristics, functions of all key components of the system
- KU6.** O&M procedures for various equipment and component
- KU7.** the maintenance schedule for all key components along with solar PV system
- KU8.** tools used in system installation and maintenance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** communicate effectively with others and various stakeholders



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- GS3.** follow organisational code of conduct
- GS4.** read and write in vernacular/English language
- GS5.** read health and safety instructions and signage
- GS6.** work constructively and collaboratively with others
- GS7.** make timely decisions for efficient utilization of resources

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>technical parameters of all key components of solar cold storage system</i>	25	25	-	-
PC1. Discuss key technical aspects of system components, their specifications and step by step process for installation including of the following :Solar PV system, Refrigeration system, Insulation type/Thermal Energy Storage system, Digital configuration of Remote monitoring system	3	-	-	-
PC2. Discuss schematic and wiring diagrams along with Check list for O&M including preventive maintenance and routine check up and system monitoring, as provided in the product manual by the manufacturer/ developer/ System Assembler.	2	-	-	-
PC3. Describe the importance of conducting testing of all solar PV components and performing fault finding and analysis, continuity checks, polarity check etc.	2	-	-	-
PC4. Discuss all key aspects of system sizing, components assembly and overall installation and O&M	2	-	-	-
PC5. Describe the DO's and Don'ts of components/system installation, operation, material handling at site, system O&M, monitoring etc	2	-	-	-
PC6. Describe the importance of Earthing for the protection of solar PV system.	2	-	-	-
PC7. Explain how to perform routine maintenance activities required for each component of the solar cold storage system	2	-	-	-
PC8. Explain how to prepare and implement preventive maintenance schedule and reactive maintenance activities.	2	-	-	-
PC9. Explain the typical faults, their causes and resolution for all system components.	2	-	-	-
PC10. Explain the de-mounting of a solar PV power plant (after commissioning).	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Discuss how to identify basic cost parameters for various solar cold storage system components, installation and O&M.	2	-	-	-
PC12. Discuss key aspects of system components warranty, Annual Maintenance Contract (AMC), service schedule, etc	2	-	-	-
PC13. Show how to analyse system components, their design attributes and specifications	-	2	-	-
PC14. Demonstrate how to choose design criteria of solar cold storage as per assessment of site conditions	-	2	-	-
PC15. Demonstrate step by step process for installation of the system and its components, including for: Solar PV system, Refrigeration system, Insulation type/Thermal Energy Storage system, Digital configuration of Remote monitoring system	-	2	-	-
PC16. Show how to analyse schematic and wiring diagrams and ensure checklist for system usage, O&M and monitoring as per the technical manual provided by the manufacturer/ developer/ System Assembler.	-	2	-	-
PC17. Analyse all key aspects of system sizing, components assembly and overall installation and O&M	-	2	-	-
PC18. Describe the DO's and Don'ts of components/system installation, operation, material handling at site, system O&M, monitoring etc.	-	2	-	-
PC19. Demonstrate how to perform earthing for the protection of solar PV system	-	2	-	-
PC20. Demonstrate how to conduct testing of all solar PV components and performing fault finding and analysis, continuity checks, polarity check etc.	-	2	-	-
PC21. Demonstrate how to carry out service and routine maintenance work for each component.	-	2	-	-
PC22. Demonstrate how to prepare and execute preventive maintenance schedule and reactive maintenance activities	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. Demonstrate how to identify typical faults, their causes and resolution for all components	-	2	-	-
PC24. Demonstrate demounting of solar PV power plant	-	1	-	-
PC25. Show how to analyse basic cost parameters for solar cold storage system components, system installation and O&M.	-	1	-	-
PC26. Show how to analyse all key aspects of availing system components warranty, performing Annual Maintenance Contract (AMC), service schedule etc	-	1	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0623
NOS Name	Technical aspects for installation, usage, performing O&M and cost of solar cold storage system
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	3
Credits	2
Version	2.0
Last Reviewed Date	12/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

SGJ/N1301: Availing low- cost financing and Case studies showcasing best practices in system installation and operation of solar cold storage system

Description

This unit provides an overview of identification and incorporation of best practices to manage the installation and operations of solar cold storage units while availing low cost financing from various sources.

Scope

The scope covers the following :

- Identify and incorporate best practices to manage the installation and operations of solar cold storage units
- Discuss how to avail low cost financing from various sources

Elements and Performance Criteria

Low-cost financing & solar cold storage best practices.

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss successful solar cold storage system installation and operation experience for various consumer segment/locations etc
- PC2.** Identify opportunities for incorporating best practices in system installation, O&M and its usage
- PC3.** Provide an overview of how to avail financial assistance/low-cost loans from government/ bank/ financial institutions towards installation of solar cold rooms
- PC4.** Demonstrate best practices of solar cold storage installation and usage
- PC5.** Analyse best practices and show how to adopt key learnings from successful case studies/ experiences in system installation, operation & maintenance, etc.
- PC6.** Demonstrate how to avail financial assistance/ low-cost financing under Government schemes, from banks/FIs etc towards installation of solar cold storage system

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** best practices of solar cold storage installation, operation & maintenance and usage
- KU2.** The concept of successful solar cold storage system installation and operation experience for various consumer segment/locations etc

Generic Skills (GS)

User/individual on the job needs to know how to:



Qualification Pack

- GS1.** complete documentation work applicable to the role
- GS2.** communicate effectively with others and various stakeholders
- GS3.** follow organisational code of conduct
- GS4.** read and write in vernacular/English language
- GS5.** read health and safety instructions and signage
- GS6.** work constructively and collaboratively with others
- GS7.** make timely decisions

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Low-cost financing & solar cold storage best practices.</i>	25	25	-	-
PC1. Discuss successful solar cold storage system installation and operation experience for various consumer segment/locations etc	10	-	-	-
PC2. Identify opportunities for incorporating best practices in system installation, O&M and its usage	8	-	-	-
PC3. Provide an overview of how to avail financial assistance/low-cost loans from government/ bank/ financial institutions towards installation of solar cold rooms	7	-	-	-
PC4. Demonstrate best practices of solar cold storage installation and usage	-	8	-	-
PC5. Analyse best practices and show how to adopt key learnings from successful case studies/ experiences in system installation, operation & maintenance, etc.	-	8	-	-
PC6. Demonstrate how to avail financial assistance/ low-cost financing under Government schemes, from banks/FIs etc towards installation of solar cold storage system	-	9	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1301
NOS Name	Availing low- cost financing and Case studies showcasing best practices in system installation and operation of solar cold storage system
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	3
Credits	1
Version	2.0
Last Reviewed Date	12/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

SGJ/N0801: Maintain Health and Safety at Solar cold storage project site

Description

This unit outlines how to ensure health and safety measures at sold storage project site

Scope

The scope covers the following :

- Ensuring Health and Safety

Elements and Performance Criteria

Ensuring Health and Safety

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the requirement and importance of ensuring personal cleanliness and hygiene.
- PC2.** Identify steps to be followed to maintain personal cleanliness while working at the project site
- PC3.** Explain the requirements for a safe work area/project site
- PC4.** Identify proper personal health and safety tools including gloves and masks
- PC5.** Identify the risks and hazards associated with solar cold storage system installations and O&M (e.g. fire, electric shocks etc)
- PC6.** Explain standard operating procedures of ensuring safety for installation of Solar cold storage solution in accordance with industry practice
- PC7.** Explain how to identify the first aid materials and administer first-aid
- PC8.** Explain how to report immediately to concerned authorities regarding sign and symptoms of illness/accidents etc
- PC9.** Discuss specified norms for waste collection and segregation for all waste categories as applicable.
- PC10.** Demonstrate how to ensure personal cleanliness and hygiene while working at project site
- PC11.** Demonstrate standard operating procedures of safety for installation of solar cold storage system in accordance with industry practice
- PC12.** Demonstrate the proper use of Gloves and Masks and other PPE
- PC13.** Demonstrate good housekeeping practices and infection control guidelines
- PC14.** Show how to mitigate the risks and hazards associated with solar cold storage system installations and O&M (e.g. fire, electric shocks etc)
- PC15.** Demonstrate how to identify the location of first aid materials and administer first-aid
- PC16.** Demonstrate safety operating procedures for installation of Solar cold storage solution
- PC17.** Demonstrate how to follow processes specified for disposal of hazardous waste, if any

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** importance of safety drills
- KU2.** importance of working in a clean and safe environment
- KU3.** organisational health and safety guidelines
- KU4.** roles and responsibilities of relevant personnel within and outside the organization
- KU5.** importance of Personal Protective Equipment required for specific job
- KU6.** waste categories and their disposal
- KU7.** housekeeping activities relevant to task
- KU8.** key internal and external source of health and safety information
- KU9.** symptoms of infection like fever, cough, swelling and inflammation
- KU10.** providing basic first aid

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete statutory documents relevant to safety and hygiene
- GS2.** interpret general safety guidelines, labels, charts and signage
- GS3.** fill safety forms and related documentation
- GS4.** communicate and collaborate with others to incorporate safety and sustainable practices
- GS5.** interpret general safety guidelines, labels, charts and signage

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensuring Health and Safety</i>	25	25	-	-
PC1. Explain the requirement and importance of ensuring personal cleanliness and hygiene.	3	-	-	-
PC2. Identify steps to be followed to maintain personal cleanliness while working at the project site	3	-	-	-
PC3. Explain the requirements for a safe work area/project site	3	-	-	-
PC4. Identify proper personal health and safety tools including gloves and masks	3	-	-	-
PC5. Identify the risks and hazards associated with solar cold storage system installations and O&M (e.g. fire, electric shocks etc)	3	-	-	-
PC6. Explain standard operating procedures of ensuring safety for installation of Solar cold storage solution in accordance with industry practice	3	-	-	-
PC7. Explain how to identify the first aid materials and administer first-aid	3	-	-	-
PC8. Explain how to report immediately to concerned authorities regarding sign and symptoms of illness/accidents etc	2	-	-	-
PC9. Discuss specified norms for waste collection and segregation for all waste categories as applicable.	2	-	-	-
PC10. Demonstrate how to ensure personal cleanliness and hygiene while working at project site	-	3	-	-
PC11. Demonstrate standard operating procedures of safety for installation of solar cold storage system in accordance with industry practice	-	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Demonstrate the proper use of Gloves and Masks and other PPE	-	3	-	-
PC13. Demonstrate good housekeeping practices and infection control guidelines	-	3	-	-
PC14. Show how to mitigate the risks and hazards associated with solar cold storage system installations and O&M (e.g. fire, electric shocks etc)	-	4	-	-
PC15. Demonstrate how to identify the location of first aid materials and administer first-aid	-	3	-	-
PC16. Demonstrate safety operating procedures for installation of Solar cold storage solution	-	3	-	-
PC17. Demonstrate how to follow processes specified for disposal of hazardous waste, if any	-	3	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0801
NOS Name	Maintain Health and Safety at Solar cold storage project site
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Health and Safety
NSQF Level	4
Credits	1
Version	2.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2026
Next Review Date	18/02/2029
NSQC Clearance Date	18/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council/Awarding Body. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS.SSC/AB will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC/AB.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4006.Basics of Solar cold storage system installation and its operation	24	26	-	-	50	14
SGJ/N0702.Site Survey and pre-requisites for installation of solar cold storage system	25	25	-	-	50	14
SGJ/N3901.Elements of solar cold storage system and their functions	25	25	-	-	50	14
SGJ/N0623.Technical aspects for installation, usage, performing O&M and cost of solar cold storage system	25	25	-	-	50	14
SGJ/N1301.Avaling low- cost financing and Case studies showcasing best practices in system installation and operation of solar cold storage system	25	25	-	-	50	14
SGJ/N0801.Maintain Health and Safety at Solar cold storage project site	25	25	-	-	50	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	15
Total	169	181	-	-	350	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.